



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCE
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO1: Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:50

Analytical Problem

Code of Conduct:

I Y. Yoganarasimha (192311254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

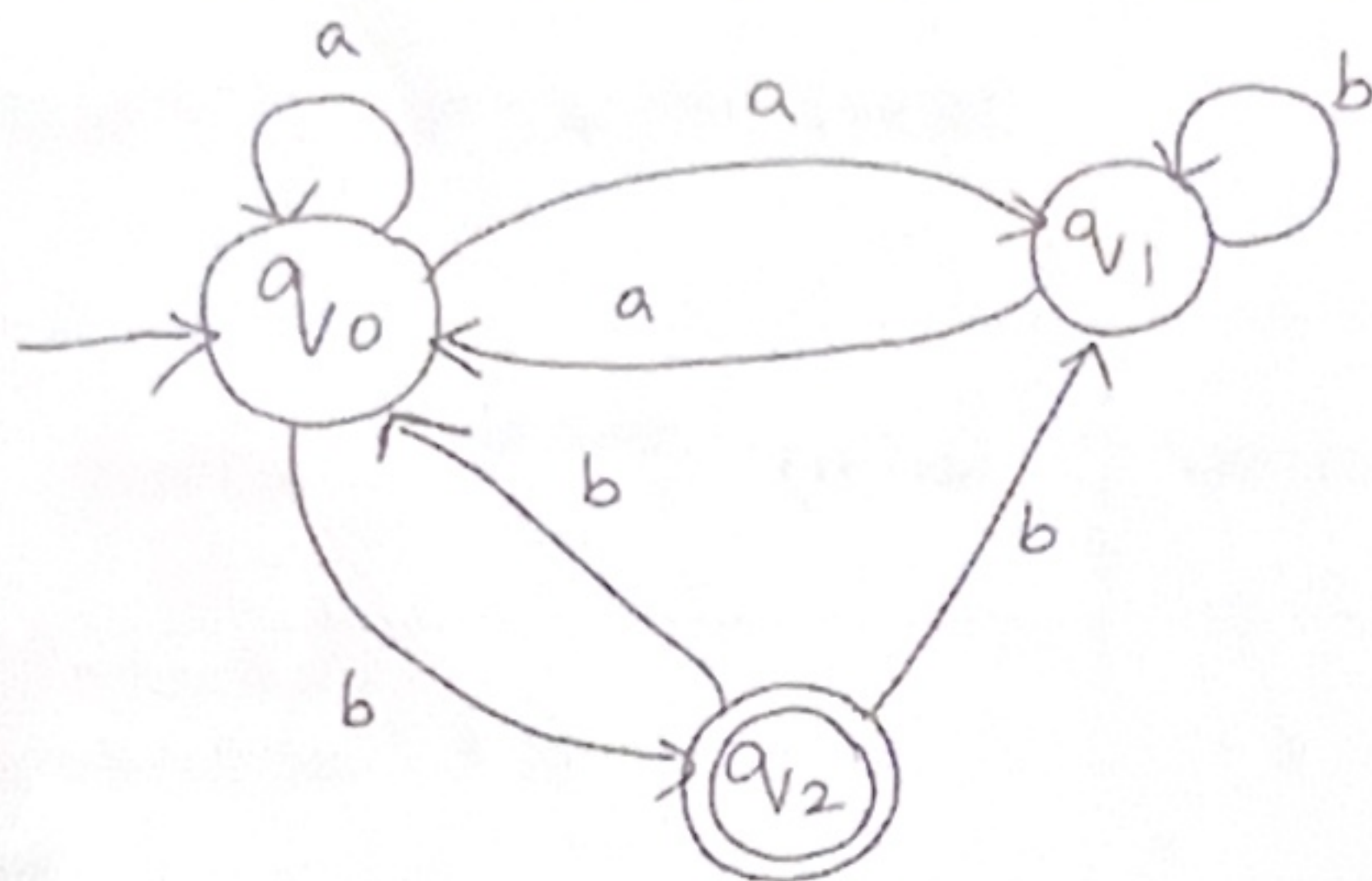
Y. Yoganarasimha



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1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L .

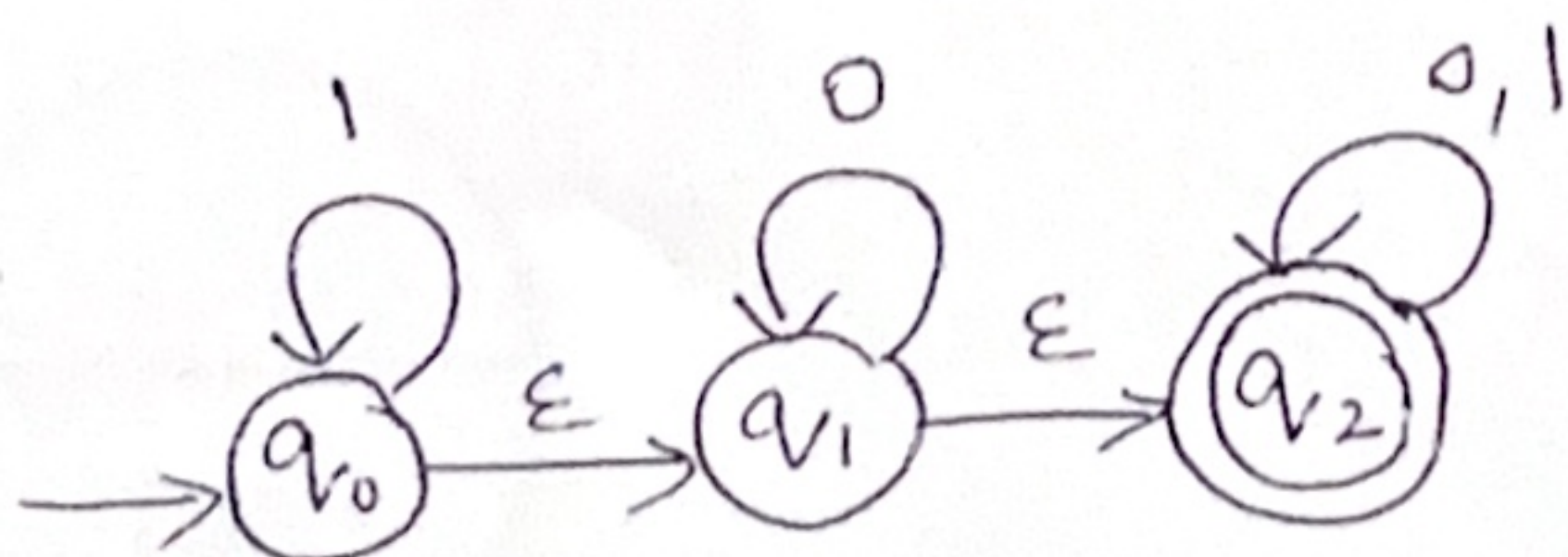


2. Let L be the language of words of length at least 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb . Construct an NFA for L . Then draw the equivalent DFA.

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA.

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

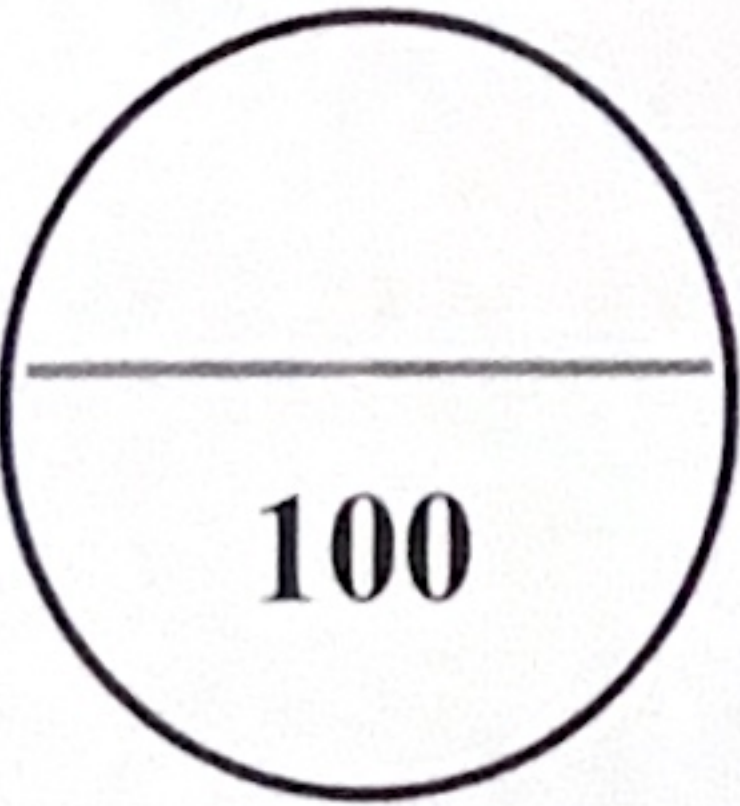
5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states.



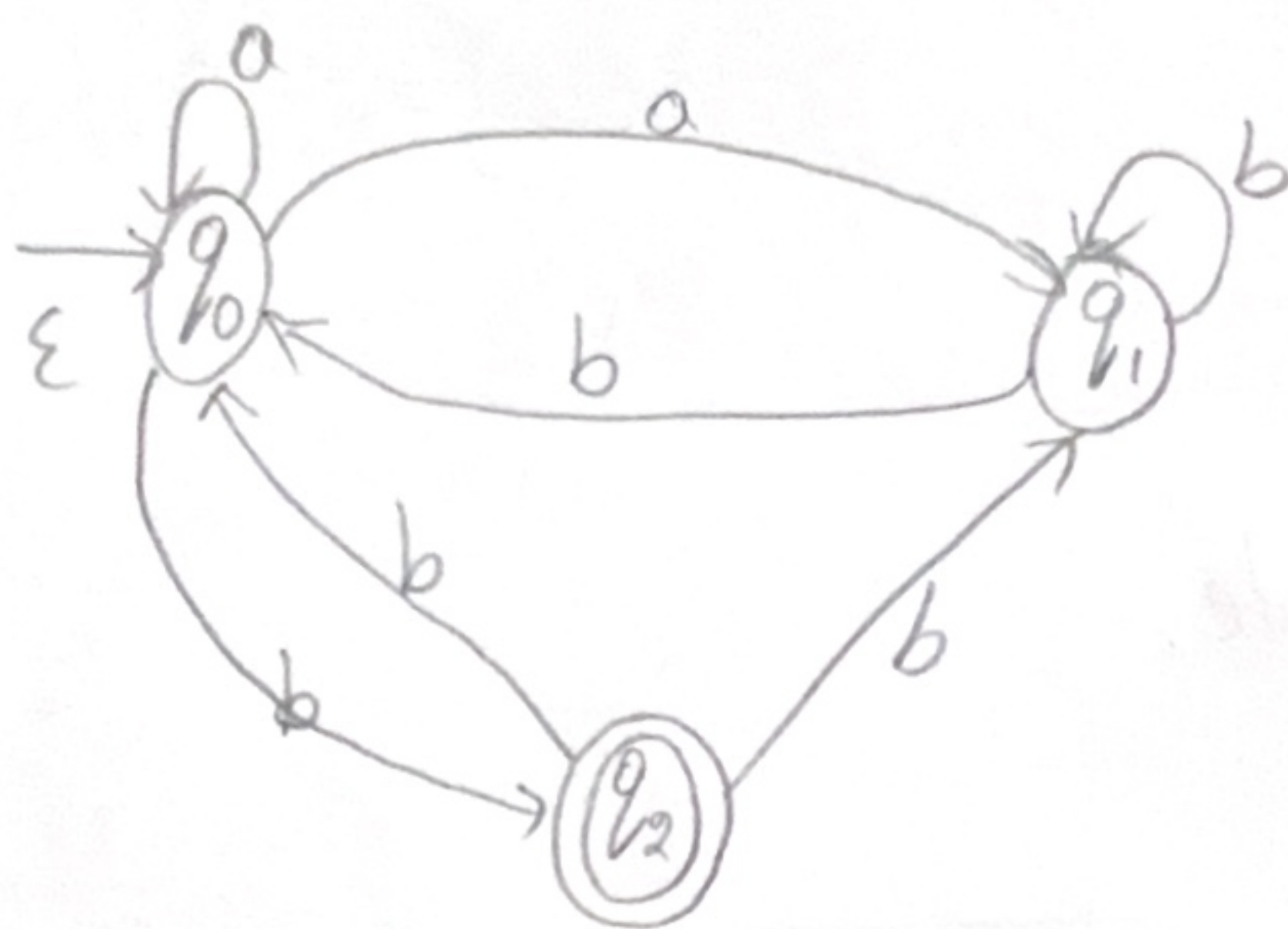


RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50



Sol:



NFA Transition Table

state	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$*q_2$	$\emptyset$	$\{q_0, q_1\}$

→ start state

* Final state

② subset construction (step-by-step)

start state = $\{q_0\}$

$$\left. \begin{array}{l} \delta(\{q_0\}, a) = \{q_0, q_1\} \\ \delta(\{q_0\}, b) = \{q_2\} \end{array} \right\} \text{From } \{q_0\}$$

$$\left. \begin{array}{l} \delta(\{q_0, q_1\}, a) = \{q_0, q_1\} \\ \delta(\{q_0, q_1\}, b) = \{q_1, q_2\} \end{array} \right\} \text{From } \{q_0, q_1\}$$

$$\left. \begin{array}{l} \delta(\{q_2\}, a) = \emptyset \\ \delta(\{q_2\}, b) = \{q_0, q_1\} \end{array} \right\} \text{From } \{q_2\}$$

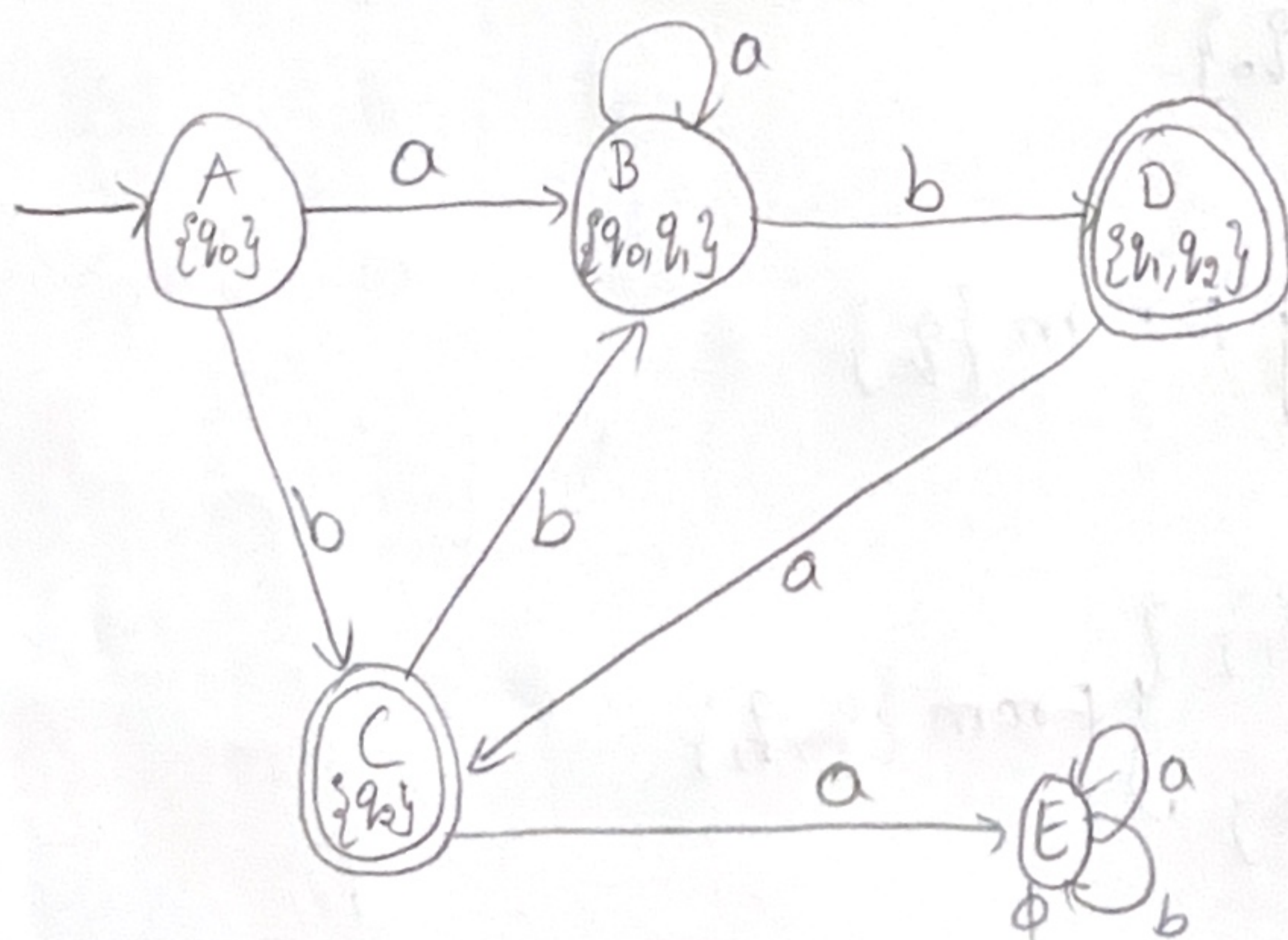
$$\left. \begin{array}{l} \delta(\phi, a) = \phi \\ \delta(\phi, b) = \phi \end{array} \right\} \text{From } \phi$$

③ Final DFA Transition Table.

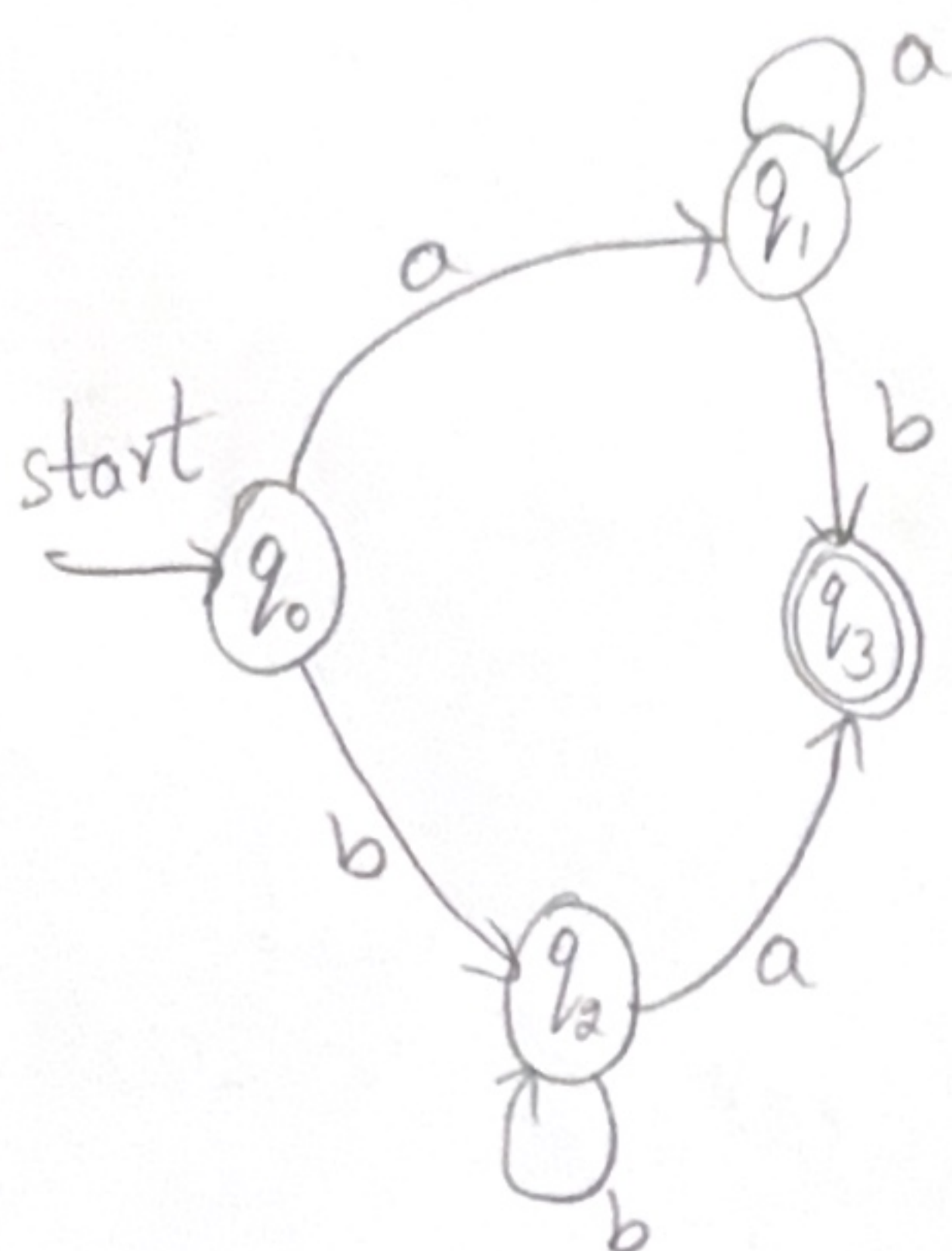
state	a	b
$\rightarrow A = \{q_0\}$	B	C
$B = \{q_0, q_1\}$	B	D
$*C = \{q_2\}$	E	B
$*D = \{q_1, q_2\}$	A	B
$E = \phi$	E	E

$A = \{q_0\}$
 $B = \{q_0, q_1\}$
 $C = \{q_2\}$
 $D = \{q_1, q_2\}$
 $E = \phi$
 start state = A
 final states = C, D

④ Final DFA



2Ans Alphabet $\Sigma = \{a, b\}$



NFA Transition Table

state	a	b
$\rightarrow q_0$	$\{q_1\}$	$\{q_2\}$
q_1	$\{q_1\}$	$\{q_3\}$
q_2	$\{q_3\}$	$\{q_2\}$
$*q_3$	$\emptyset$	$\emptyset$

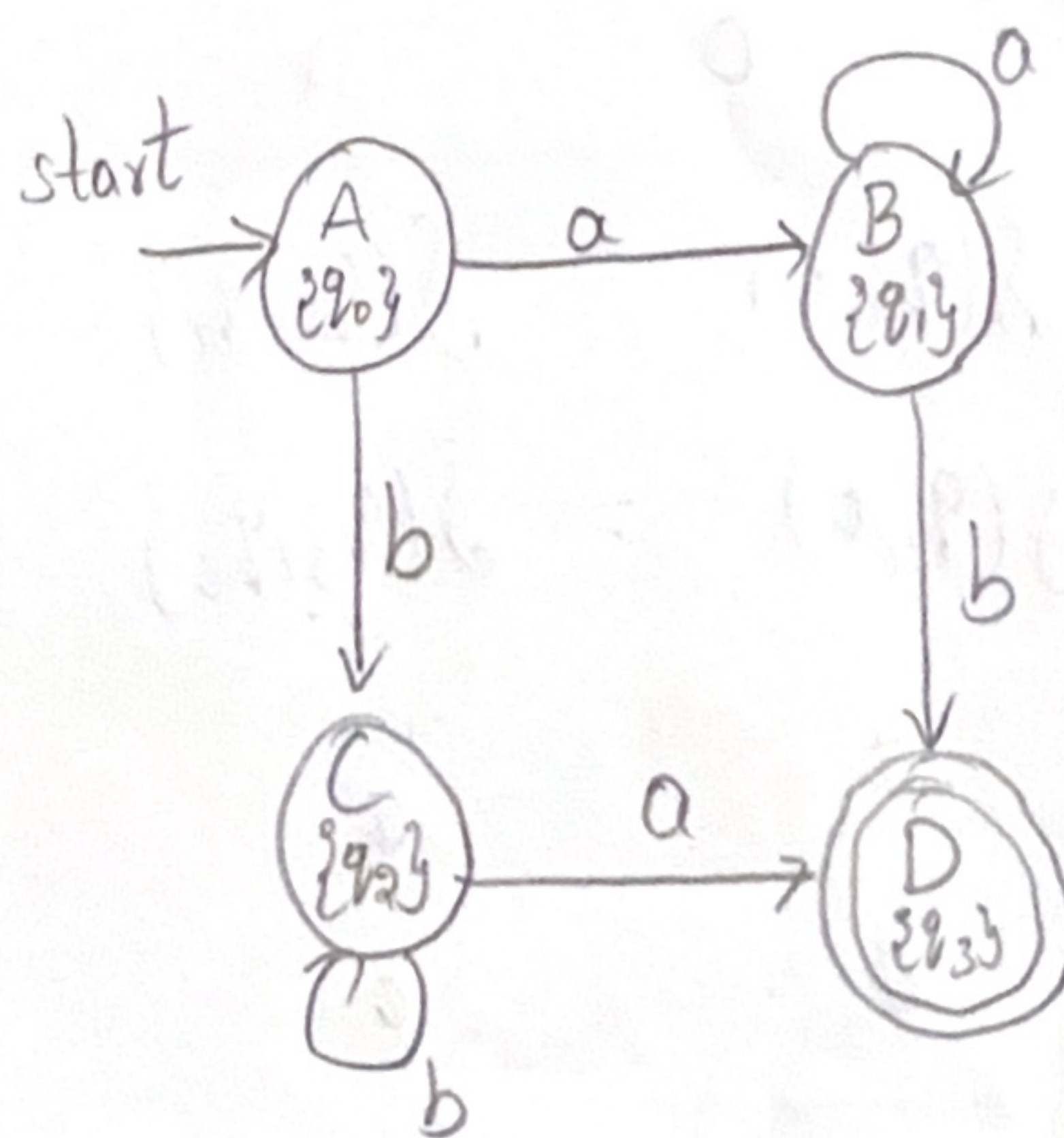
$\rightarrow$ start state : q_0

* Accepting state : q_3

Equivalent DFA (using subset construction)

DFA Transition Table.

DFA	a	b
$\rightarrow A = \{q_0\}$	B	C
$B = \{q_1\}$	B	D
$C = \{q_2\}$	D	C
$*D = \{q_3\}$	$\emptyset$	$\emptyset$



state mapping

$A = \{q_0\}$

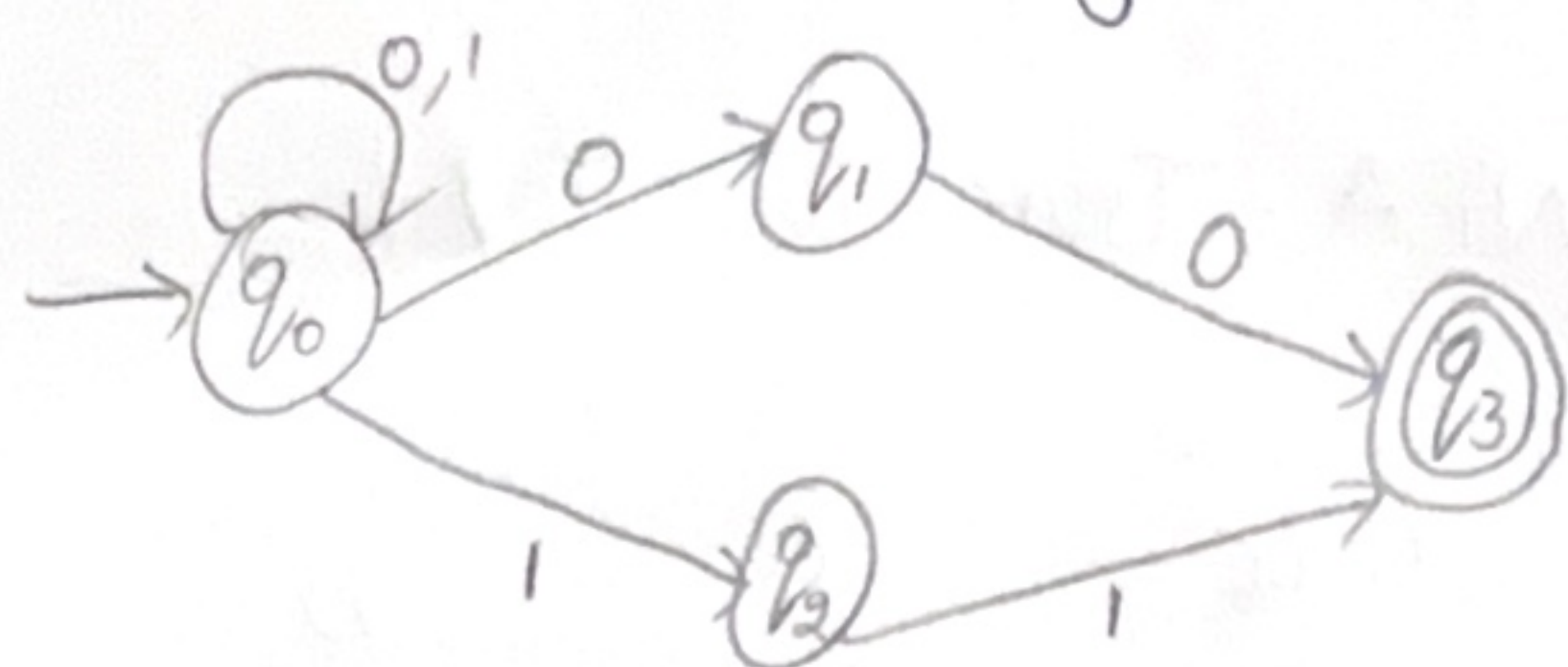
$B = \{q_1\}$

$C = \{q_2\}$

$D = \{q_3\}$

$\emptyset$ = dead state

3A. NFA State Diagram.



② Formal Definition of the NFA

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\}$$

$$q_0 = q_0$$

$$F = \{q_3\}$$

δ	0	1
q_0	$\delta(q_0, 0) = \delta(q_0, q_1)$	
q_1	$\delta(q_1, 0) = \delta(q_3, q_3)$	
q_2	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$

③ String Acceptance Verification: 01011

Initial state: $\{q_0\}$

$$\text{Read 0: } \delta(q_0, 0) = \{q_0, q_1\}$$

$$\begin{aligned} \text{Read 1: } \delta(\{q_0, q_1\}, 1) &= \delta(q_0, 1) \cup \delta(q_1, 1) \\ &= \{q_0, q_2\} \cup \emptyset = \{q_0, q_2\} \end{aligned}$$

3.

$$\text{Read 0: } \delta(\{q_0, q_2\}, 0) = \delta(q_0, 0) \cup \delta(q_2, 0) \\ = \{q_0, q_1\} \cup \emptyset = \{q_0, q_1\}$$

↓

$$\text{Read 1: } \delta(\{q_0, q_1\}, 1) = \delta(q_0, 1) \cup \delta(q_1, 1) \\ = \{q_0, q_2\} \cup \emptyset = \{q_0, q_2\}$$

↓

$$\text{Read 1: } \delta(\{q_0, q_2\}, 1) = \delta(q_0, 1) \cup \delta(q_2, 1) \\ = \{q_0, q_2\} \cup \{q_3\} = \{q_0, q_2, q_3\}$$

Final states contain q_3 , so 01011 is accepted

The Specific accepting path is

$$q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_2 \rightarrow q_3.$$

— X —

4A.

Equivalent DFA

states:

$q_0 \Rightarrow \text{start}$

$q_1 \rightarrow \text{matched}(0)$

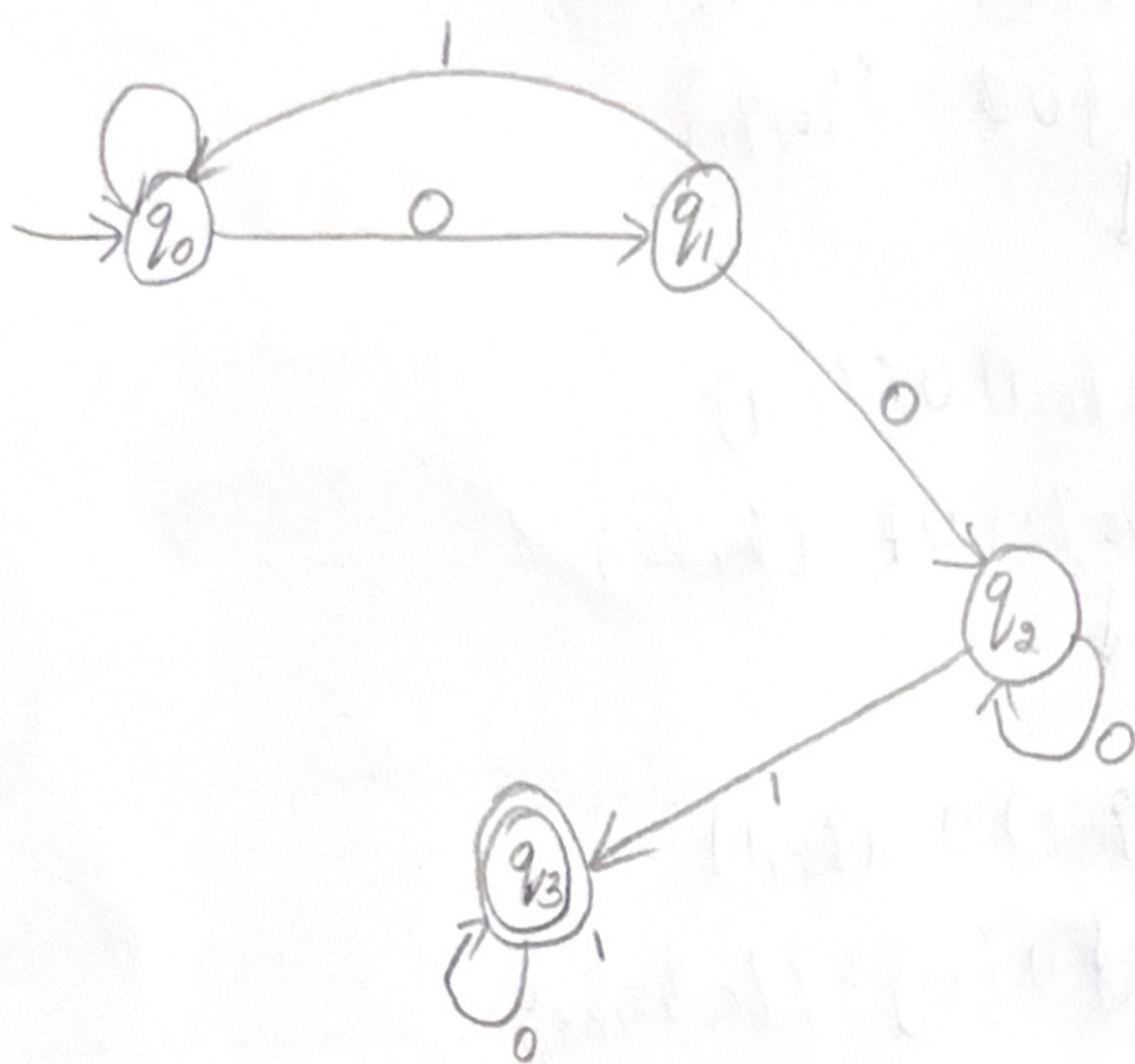
$q_2 \rightarrow (00)$

$q_3 \rightarrow (001)$

Transition Table.

state	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
$*q_3$	q_3	q_3

DFA Diagram.



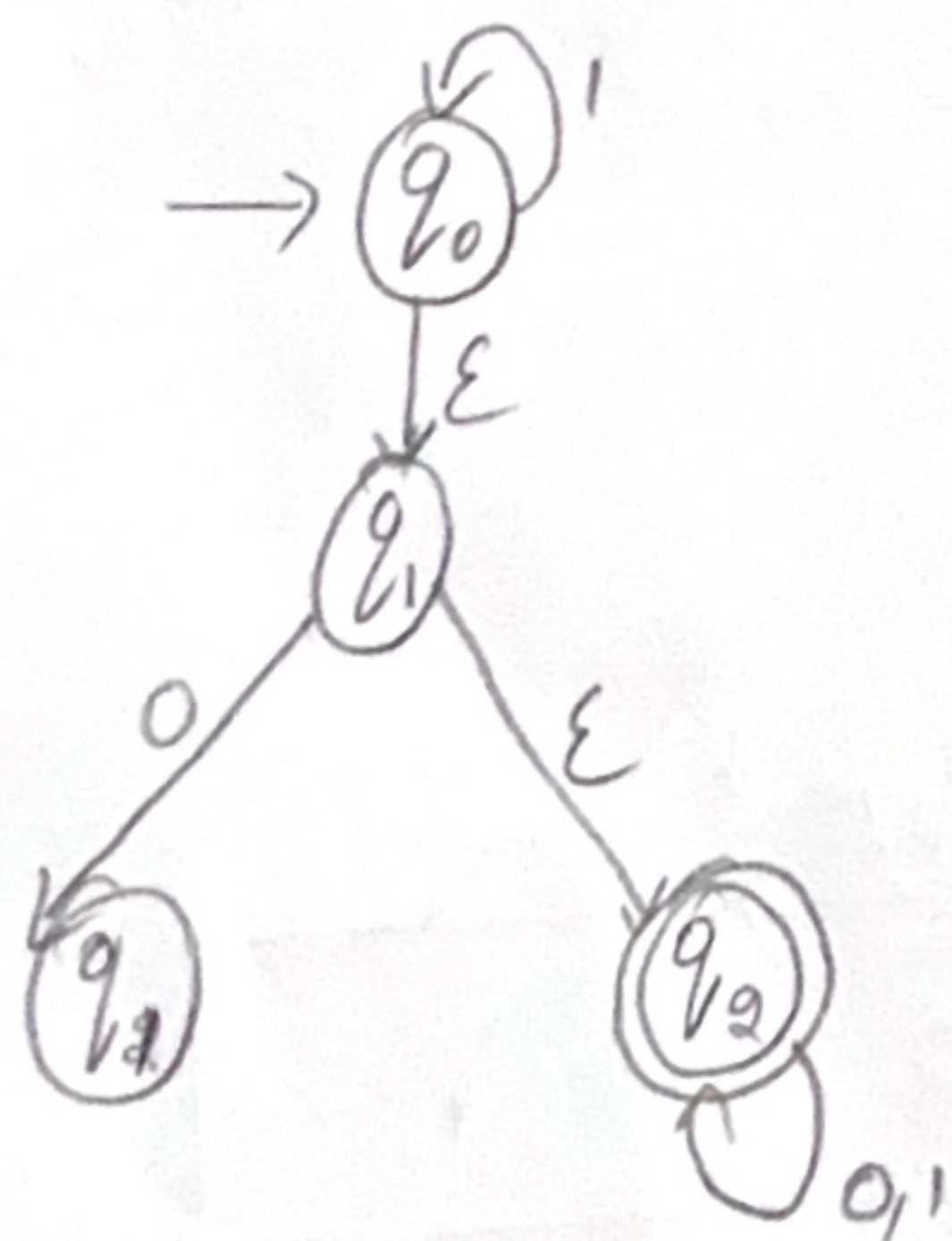
∴ Final state: *q₃.

— X —

5A.



Given ε-NFA



start state = (q₀)

final state = q₃

Find ε-closures : (q₀)

① q₀ → ε q₁ → ε q₂

ε-closure(q₀) = (q₀, q₁, q₂)

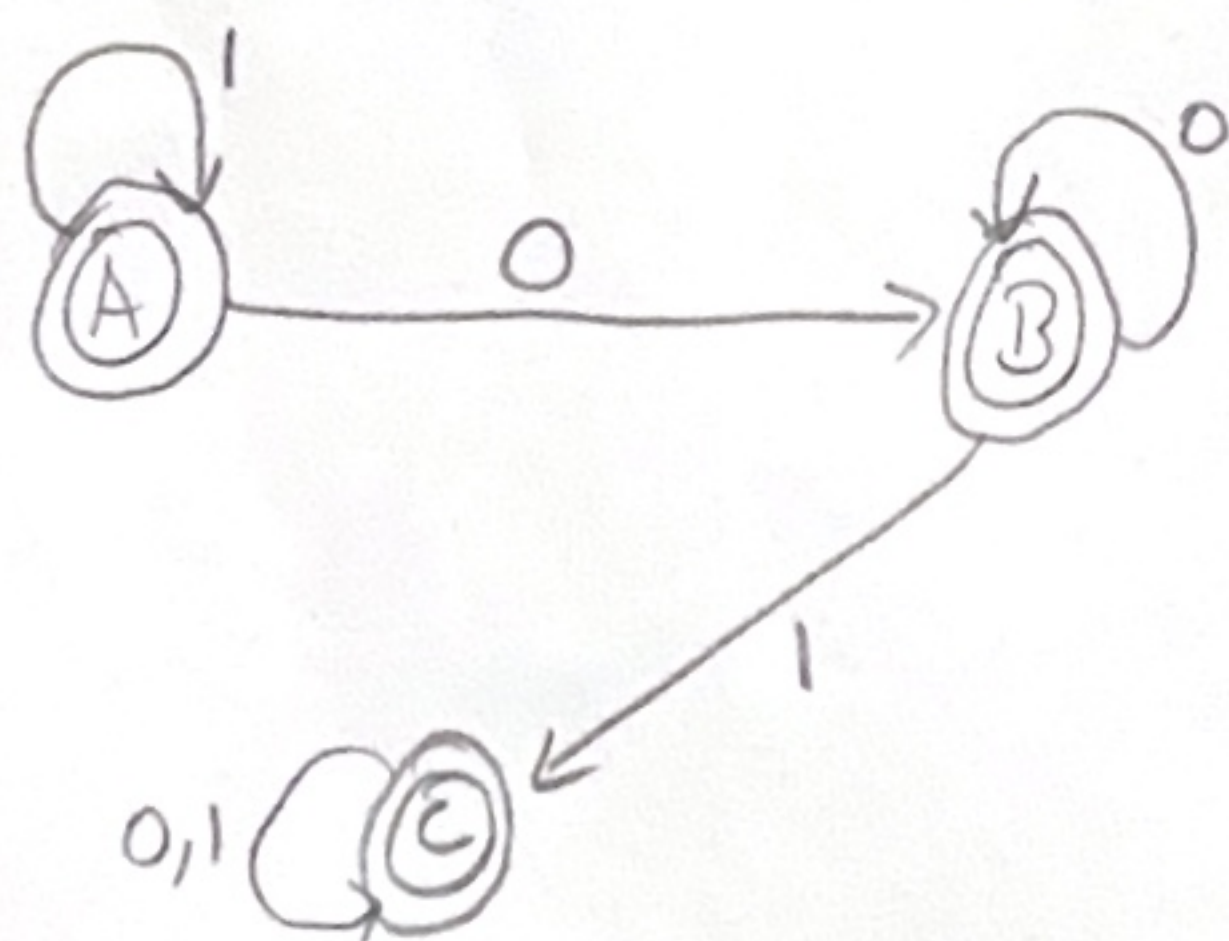
② $q_1 \rightarrow \epsilon q_2$
 $\epsilon\text{-closure}(q_1) = (q_1, q_2)$

③ $\epsilon\text{-closure}(q_2)$
 NO ϵ -transition; so;
 $\epsilon\text{-closure}(q_2) = (q_2)$

DFA Transition Table:-

DFA states	0	1
$\rightarrow A^* = (q_0, q_1)$	B	A
$*B = (q_1, q_2)$	B	C
$*C = \{q_2\}$	C	C

Equivalent DFA Diagram



$\therefore$ Initial and Final states:.

Initial state = (A) = $\{q_0, q_1, q_2\}$

Final states = A, B, C

$\therefore$ Because A, B, C all states here contains q_2 .